

Worked Solutions & Explanations:

1 B

$$\langle \rho \rangle = (1002 + 998 + 997 + 1001 + 999)/5 = 999.4 \text{ kg m}^{-3}, \text{ closer to } 1000 \text{ kg m}^{-3}$$

$$\text{difference from true value} = 1000 - 999.4 = 0.6 \text{ kg m}^{-3}$$

$$\Delta \rho = \rho_{\text{max}} - \rho_{\text{avg}} = 1002 - 997 = 5 \text{ kg m}^{-3}$$

2 D